Grinding at the Edge: Empirical Signer-Cost Distributions for Bitcoin-Tuned Hash-Based Signatures

Georgios Nanos · DRAFT 2026-10-24 · under review; open items are listed in §7

> **Draft status.** Every number in this draft comes from `analysis/out/` (run `196e7e91e907`) or from `kb/claims/all_claims.yaml`. Signature sizes use R = 16 B (FIPS 205, SPHINCS-Parameters HEAD); Kudinov–Nick's tables use R = 32 B and are 16 B larger.

Abstract

Hash-based signatures are the conservative post-quantum option for Bitcoin. Their size can be cut by *grinding*: the signer retries a hash until a condition holds, so checksum chains (WOTS+C) or FORS tree paths (FORS+C, PORS+FP) can be dropped from the signature. Prior work tabulates the expected search cost and an analytic 2^{-30} worst case in abstract hash calls.

We measure the cost. On a 45 W laptop GPU we ran 11.16 million reference-verified grinds across 37 grind configurations, which serve 40 parameter sets at 2^{20}, 2^{30} and 2^{40} signature limits. We cross-checked them on a SHA-NI laptop CPU. Results:

- Every measured mean is within 2 % of the analytic expectation, and the distributions are geometric down to the 10^{-4}…10^{-6} tail.
- The widely used cost model under-prices FORS+C and PORS+FP attempts by 1.5–1.75× under FIPS 205's SHA-2 instantiation, and over-prices the counter-in-H_msg variant by 1.33×.
- Grinding is at most 18 % of total signing work; tree building dominates.
- A signer budget of 10 s at p99 on a laptop CPU, or under one second on a laptop GPU, buys 596–1,052 bytes per signature over a hardware-wallet-safe set: 17–23 % of a 1-input/2-output transaction.
- Verification cost never increases.

We release the CUDA/NVRTC grinder, the dataset and a one-command reproduction.

1. Introduction

BIP-360 (P2MR) was merged on 2026-02-11 and leaves the post-quantum signature opcode to a future BIP. Kudinov and Nick's Bitcoin-tuned SPHINCS+ variants (ePrint 2025/2203) show that two levers shrink hash-based signatures:

- **fewer signatures per key:** 2^{20}–2^{40} instead of 2^{64};
- **grinding:** a pure work-for-bytes trade made at signing time.

How much work a signer can afford is a free parameter of the design, and the community disagrees about it. Positions range from "seconds per input on low-power devices is fine" (Maxwell, RQ-02) to "a couple of minutes is not a big deal" (Blockstream, RQ-05). SHRINCS fixes a single parameter set (RQ-41).

Kudinov and Nick opened the bitcoindev thread with the question **"What are the concrete performance requirements across various hardware, including low-power devices?"** (OQ-1). Maxwell's answer to whether several parameter sets should be standardized was **"it would depend on their relative performance"** (OQ-2).

This paper supplies that relative performance, measured. Contributions:

1. **Validation (RQ1).** The analytic expected cost holds to 2 % for WOTS+C, FORS+C (two layouts) and PORS+FP, and the full distribution is geometric.
2. **Per-attempt pricing (H1b).** Counted under FIPS 205 SHA-2, grinding attempts cost 1 / 6 / 3 / 7 SHA-256 compressions (WOTS+C / FORS+C R-grind / FORS+C ctr-grind / PORS+FP), not 1 / 4 / 4 / 4.
3. **Latency per signer class (RQ2, OQ-1).** p50 / p99 / p99.9 signing time from hardware wallets to GPUs, measured on two classes and scaled from cited rates for four.
4. **Frontier and fees (RQ3/RQ4, OQ-2).** The smallest signature per time budget and per class, and its fee effect.
5. **Open artifacts:** kernels, data, notebooks, reproduction.

2. Background

SPHINCS+ / SLH-DSA. A hypertree of d layers of XMSS trees, each signing the root below with WOTS, over a few-time signature (FORS) on the message digest. FIPS 205 fixes the SHA-2 instantiation:

- H_msg = MGF1-SHA-256(R ‖ PK.seed ‖ SHA-256(R ‖ PK.seed ‖ PK.root ‖ M), m);
- PRF_msg = HMAC-SHA-256.

WOTS+C. The signer grinds a counter until the base-w digits of d = Th(PK.seed, ADRS, m ‖ count) sum to a fixed target S. This removes the checksum chains. Success probability p = v / w^len1. The signature carries the counter.

FORS+C. The signer grinds until the last FORS tree opens leaf 0, which happens with $p = 2^{-a}$, so that tree's path is dropped. Kudinov–Nick put the counter into PRF_msg ("R-grind": fresh R per attempt, nothing extra stored). The original SPHINCS+C appends a counter to H_msg ("ctr-grind": R fixed, 4-byte counter stored).

PORS+FP. PORS with forced pruning. The signer grinds R until the k distinct indices derived from H_msg have an Octopus authentication set of size ≤ m_max. Expected work comes from Abri–Katz's exact PMF. With a 512-bit H_msg, an attempt can also fail to yield k distinct indices; we include that probability (P_enough, 0.987–1.000 for the reference rows).

Tail model. Independent attempts make the number of tries T geometric: P(T > t) = (1 − p)^t. So p99 ≈ 4.61/p, p99.9 ≈ 6.91/p and the 2^{-30} quantile ≈ 20.8/p.

3. Methodology

Parameter points. We use 40 shortlisted points:

- the 29 Kudinov–Nick table rows (reproduced exactly by our port of `costs.sage`: size, verification cost, expected search and 2^{-30} worst case);
- per signature limit, a **hw_wallet_safe** point (every grind ≤ 2^{12} tries), a **gpu_signer_moderate** point and a **gpu_signer_max** point;
- two stress points.

The search space was capped at grinds ≤ 2^{24} expected tries and signing ≤ 2^{30} compressions. The grind distribution depends only on (w, S, z), on (k, a) and the layout, or on (k, a, h, m_max), so the 40 points need 37 grind configurations.

Byte layouts. The authors give no byte-level SHA-256 spec for FORS+C or PORS+FP, so we fixed FIPS 205-faithful layouts (`kb/constructions/`):

- WOTS+C: ADRSc (type 7) ‖ m ‖ count_{32} in one block after the PK.seed midstate;
- FORS+C: last a-bit chunk of md is zero;
- R-grind: HMAC message opt_rand ‖ M ‖ i_{32}.

Tries do not depend on these choices; per-attempt cost and test vectors do.

Kernels. NVRTC-compiled CUDA (no toolkit needed), one kernel per layout. Each thread strides counters, and the lowest successful counter wins (atomicMin), so the result is deterministic and equals a sequential search. Many independent instances are batched per launch, and each instance is one sample.

Correctness.

- **Gate.** A campaign starts only after the E1 gate passes: 10 configurations at reduced difficulty (all acceptance paths, including FORS masks straddling 32-bit words and non-power-of-two PORS t), 10^4 instances each. Every counter matches the pure-Python reference with a full minimality check.
- **Campaign.** All 11.16 M campaign counters were re-checked by the reference.
- **CPU cross-check.** A separate SHA-NI CPU grinder reproduced all WOTS+C counters bit for bit.

Samples. 10^6 per WOTS+C config, $2{\cdot}10^5$ per FORS+C config, 10^5 per PORS+FP config. The three PORS+FP configs with E[tries] ≈ $1.2{\cdot}10^7$ got $2{\cdot}10^4$.

Hardware. MSI Thin 15 B13VE:

- RTX 4050 Laptop GPU at its 45 W TGP floor (20 SMs, sm_89). E0 sustained single-block SHA-256: $2.15{\cdot}10^9$ compressions/s.
- i5-13420H (4P+4E, SHA-NI). E5 sustained, 12 threads: $2.33{\cdot}10^8$ compressions/s; 1 thread: $2.93{\cdot}10^7$.

Board power, clocks and clock-event reasons were logged at 10 Hz. The GPU sat at its power cap; 0.014 % of busy samples were flagged by any slowdown reason.

Signer classes. For the other classes we use cited typical rates (`kb/hardware.yaml`), labelled as cited/derived:

- Cortex-M4 software SHA: $4{\cdot}10^4$/s;
- MCU with SHA peripheral: $1.2{\cdot}10^5$/s;
- Raspberry Pi 5: $7.4{\cdot}10^7$/s;

- 8× RTX 4090: $1.72 \cdot 10^{11}$/s.

Full signing cost (E4) = the model's non-grinding work (tree building) + the measured per-signature grind distribution (d WOTS+C layers + one FTS grind, resampled), priced at the counted per-attempt cost.

Statistics. Mean ratio with 95 % CI. A randomized probability integral transform against the *analytic* p (nothing fitted), tested with KS (Holm-corrected) and Anderson–Darling. Σz^2 across configs.

4. Results

4.1 Validation (RQ1, H1)

	value
configs / samples	37 / 11.16 M, all reference-verified
max |mean/analytic − 1|	1.66 % (`pors-k9-a13-h48-m77`, n = $2 \cdot 10^4$, z = −2.36); rerun at n = 10^5: 0.08 %, z ≈ −0.26
Σz^2 (37 dof)	39.6, p = 0.36; 2 configs outside their 95 % CI (1.9 expected)
KS on PIT, Holm 5 %	0 rejections
AD on PIT, 5 %	2 rejections (1.9 expected)
variance / geometric variance	0.963–1.026 (median 1.000)
p99 / mean	4.56–4.65 (theory 4.61)
p99.9 / mean	6.73–7.05 (theory 6.91)

All configs collapse onto e^(−x) when tries are divided by E[tries] (Fig. 1). The analytic model, including the P_enough correction for PORS+FP, is accurate for the mean and the shape.

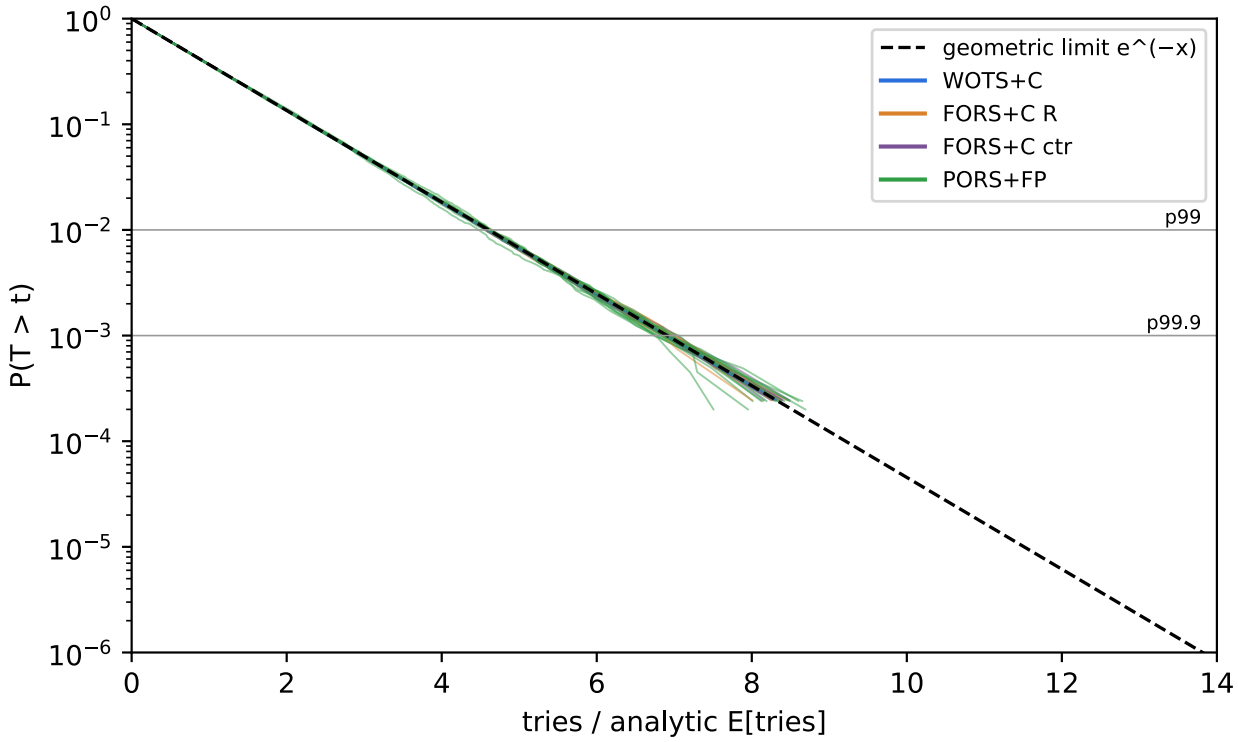


Figure 1. Survival function of tries divided by E[tries] for all 37 configurations; every curve follows e^(−x).

4.2 What an attempt really costs (H1b)

grind	model (costs.sage)	counted (FIPS 205 SHA-2)	GPU attempts/s	useful comp/s ÷ E0
WOTS+C	1	1	$1.35 \cdot 10^9$	0.63
FORS+C R-grind	4	6	$3.49 \cdot 10^8$	0.97
FORS+C ctr-grind	4	3	$7.25 \cdot 10^8$	1.01
PORS+FP	4	7	$1.81 \cdot 10^8$	0.59

The model assumes a single-SHA-256 H_msg. FIPS 205's H_msg is nested (inner hash + MGF1) and PRF_msg is HMAC. Two consequences:

- **Layout choice.** On the GPU, ctr-grind runs 2.08× more attempts per second than R-grind but stores a 4-byte counter. This is a measurable design choice, not just a 4-byte question.
- **Kernel efficiency.** WOTS+C and PORS+FP kernels run at about 60 % of raw SHA throughput because of per-attempt digit and Octopus work. FORS+C runs at raw speed.

Re-priced full signing cost differs from the model by −1 % … +8 %.

4.3 Grinding is a minor part of signing

Mean grinding is 0–18 % of total signing work across all 49 point × layout rows; tree building dominates. Even for the gpu_signer_max points (≈ $5–8 \cdot 10^8$ compressions), the base cost is 86–90 % of the total. Two consequences:

- the grind tail moves the full-signing p99 only slightly;

- the cost of the small "GPU-signer" sets is mainly larger hypertree layers (h/d = 16), not grinding.

Per signature, p99.9/mean of the grind ranges from 2.96 (several comparable WOTS+C layers, which concentrate) to 7.1 (one dominant FTS grind). Kudinov–Nick's 2^{-30} worst case is 6.3–20.9× the mean.

4.4 Latency per signer class (RQ2, OQ-1)

p99 full signing time = model base + measured grind, divided by the class rate:

limit	profile	bytes	Cortex-M4 [c]	MCU+SHA [c]	Pi 5 [c]	i5 SHA-NI [m]	RTX 4050 [m]
2^{20}	hw_wallet_safe	2876	2.7 min	54 s	88 ms	28 ms	3.0 ms
2^{20}	gpu_signer_moderate	2504	24 min	8.0 min	0.78 s	0.25 s	27 ms
2^{20}	gpu_signer_max	2280	6.1 h	2.0 h	12 s	3.8 s	0.41 s
2^{30}	hw_wallet_safe	3456	2.2 min	43 s	70 ms	22 ms	2.4 ms
2^{30}	gpu_signer_moderate	2908	20 min	6.7 min	0.65 s	0.21 s	22 ms
2^{30}	gpu_signer_max	2664	6.3 h	2.1 h	12 s	3.9 s	0.42 s
2^{40}	hw_wallet_safe	3944	2.8 min	56 s	90 ms	29 ms	3.1 ms
2^{40}	gpu_signer_moderate	3216	35 min	12 min	1.1 s	0.36 s	39 ms
2^{40}	gpu_signer_max	2892	8.0 h	2.7 h	16 s	5.0 s	0.54 s

[c] cited/derived rate · [m] measured on the test laptop. Times assume perfect parallelism within a class (§7).

4.5 Frontier (RQ3, H2, OQ-2)

Fig. 2 plots bytes against p99 signing work per limit. Smallest shortlisted signature within a p99 budget:

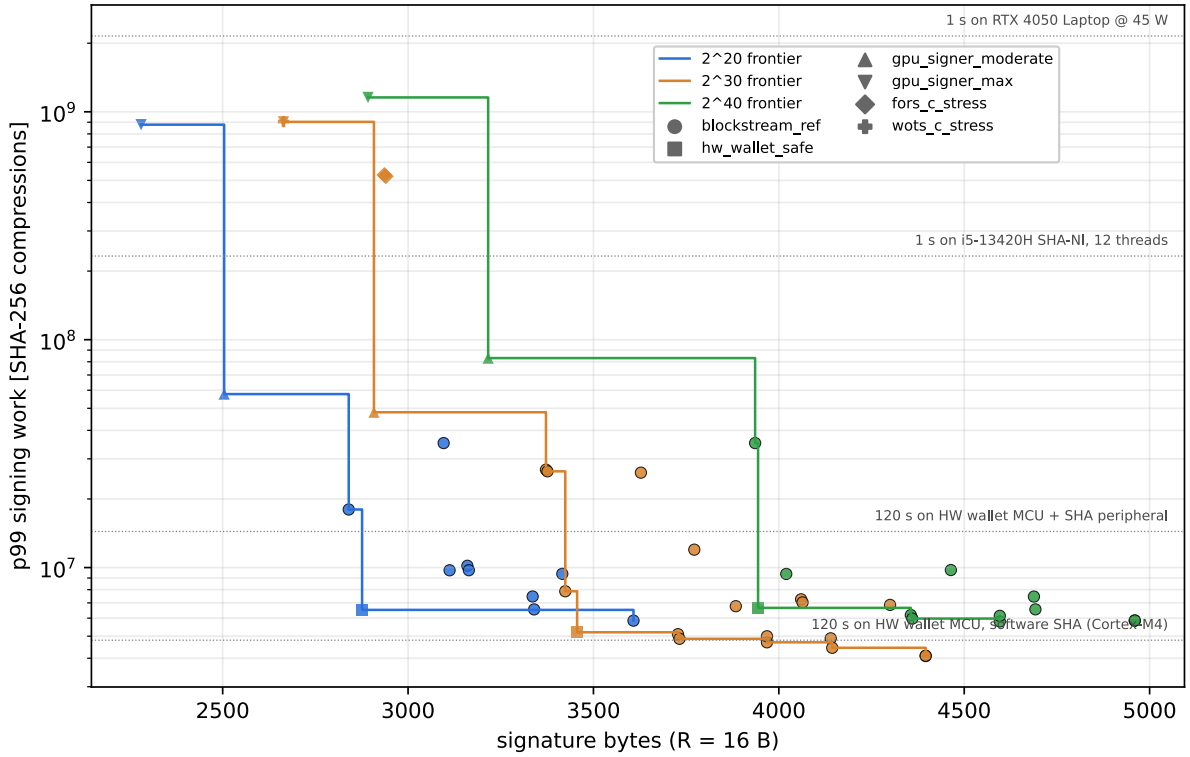


Figure 2. Signature bytes against p99 signing work (SHA-256 compressions) per signature limit, with the Pareto frontier and one-second and two-minute budgets of four signer classes.

limit	Cortex-M4, 120 s	MCU+SHA, 120 s	laptop CPU, 1 s	laptop CPU, 10 s	laptop GPU, 1 s
2^{20}	none ≤ 120 s	2876	2504	2280	2280
2^{30}	3968	3424	2908	2664	2664
2^{40}	none ≤ 120 s	3944	3216	2892	2892

The GPU reaches the smallest shortlisted point at every budget. Because the shortlist caps signing at 2^{30} compressions, the true GPU frontier lies further left, and the GPU savings are lower bounds.

4.6 Fees (RQ4)

Per input, against the hw_wallet_safe point of the same limit (BIP-360 P2MR, depth 1, 32-byte PQ key):

limit	profile	Δbytes	ΔvB	Δsat @ 2 / 10 / 50 sat/vB	share of a 1-in/2-out tx
2^{20}	gpu_signer_moderate	−372	−93	−186 / −930 / −4,650	−10.6 %
2^{20}	gpu_signer_max	−596	−149	−298 / −1,490 / −7,450	−17.0 %
2^{30}	gpu_signer_moderate	−548	−137	−274 / −1,370 / −6,850	−13.4 %
2^{30}	gpu_signer_max	−792	−198	−396 / −1,980 / −9,900	−19.4 %
2^{40}	gpu_signer_moderate	−728	−182	−364 / −1,820 / −9,100	−15.9 %
2^{40}	gpu_signer_max	−1,052	−263	−526 / −2,630 / −13,150	−23.0 %

4.7 Verification (H3) and energy (RQ5)

- **Verification.** In all 15 pairs that differ only in FTS compression, verification compressions fall by 15–40. The gpu_signer points verify cheaper than the hw_wallet_safe points of the same limit (e.g. 2^{40}: 5,893 vs. 10,026 compressions) because they have fewer, taller layers.
- **Energy.** Measured GPU energy is 20.8 nJ per raw compression. Per attempt: WOTS+C 27 nJ, ctr-grind 59 nJ, R-grind 124 nJ, PORS+FP 241 nJ. The largest GPU-signer signature costs about 18 J on the RTX 4050. Energy is not a constraint for desktop or GPU signers.

5. Discussion

Design for 7× the mean, not the mean. The tails are textbook geometric. A wallet UI should budget about 7× the mean grind (p99.9) and expose progress, not a fixed timeout at the mean. Because tree building dominates, full signing time is much tighter than the grind tail; per-signature p99.9/mean of grinding alone is 3–7.

Two signer profiles are a real choice. Between a hardware-wallet-safe and a GPU-signer set the gap is 596–1,052 B (17–23 % of a simple spend) at the limits studied. Even a laptop CPU at a 1 s budget gets 372–728 B of it. Maxwell's condition for more than one parameter set ("relative performance") is therefore met with a large margin. Free configuration hurts privacy (RQ-40), which argues for exactly two named profiles rather than a menu.

Price attempts at FIPS 205 cost. Cost tables that feed parameter selection should use 6 (R-grind) or 7 (PORS+FP) compressions per attempt, or state the non-FIPS H_msg they assume. For signers that store a counter anyway, ctr-grind halves FORS+C grinding.

Overgrinding (OQ-3). Raising the WOTS+C target S trades signer work for verifier work at fixed size. At w = 4, S = 96 → 128 removes a third of per-layer chain steps for about 650× more grinding. Measured (run `oq3-eb65c6cd81a0`, 21 configs, 10^5–10^6 samples each, notebook 05, Fig. 3): every mean is within 0.26 % of analytic.

- **w = 4.** S = 128 costs $1.5{\cdot}10^4$ tries/layer, a p99 of 1.7 s on a software-SHA Cortex-M4. S = 140 (−46 % chain steps) costs $6.2{\cdot}10^6$ tries, a p99 of 12 min on that MCU but 0.12 s on a 12-thread laptop CPU.
- **w = 16.** S = 240 → 304 removes 27 % of chain steps for 20× more tries, raising the MCU p99 from 7.5 ms to 0.15 s.

Overgrinding is cheap when the signer is a PC. For a software-SHA hardware wallet at w = 4, the p99 passes 1 s once about a third of the chain steps are removed (S ≥ 128).

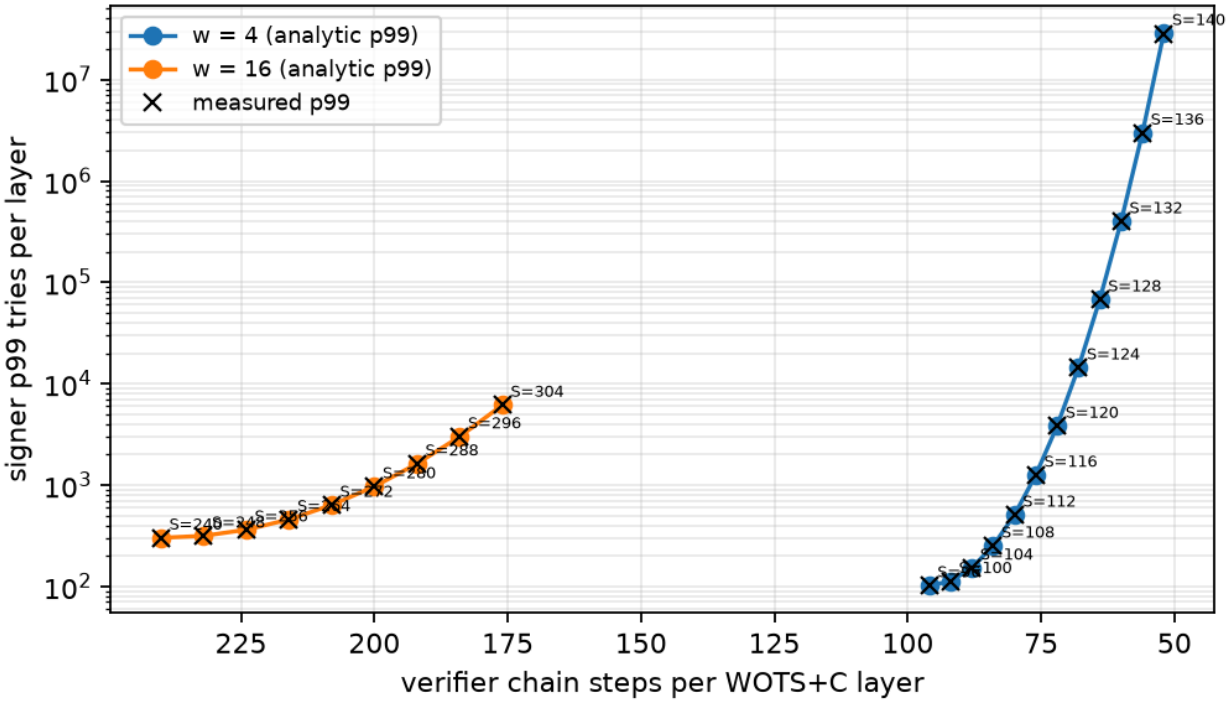


Figure 3. OQ-3 overgrinding: signer p99 tries per layer against verifier chain steps, analytic (lines) and measured (crosses).

6. Related work

- Kudinov–Nick 2025 (ePrint 2025/2203): parameters, analytic expected and 2^{-30} costs; our baseline.
- Hülsing–Kudinov–Ronen–Yogev, SPHINCS+C: WOTS+C, FORS+C.
- Abri–Katz: PORS+FP.
- Fluhrer–Dang 2024/018: smaller SPHINCS+ without grinding; grinding buys ≈ 300 B at equal signing budget.
- The SHRINCS BIP draft: stateful + plain-SLH-DSA stateless path.
- Blockstream's 2026 hardware-wallet benchmarks: means on real devices; our tails complement them.

7. Threats to validity

Summarized from `kb/threats_to_validity.md`:

- **z = 1 security (open).** The z = 1 (forced zero chain) gpu_signer points have no analysed security impact in our model. They are reported as measured, not endorsed. SPHINCS+C's theorems are stated for general z with no z-dependent bound term. What remains is independent review and the verifier rule that the zeroed digit must be checked (threats §1.2).
- **Our byte layouts.** Tries transfer to any author layout; per-attempt cost is reported for both readings of H_msg.
- **Modelled tree-building cost.** The base cost is modelled, not measured.
- **Cited ladder.** Four of six signer classes use cited rates; tries remain the primary, hardware-independent metric.

- **Parallelism assumption.** Throughput is assumed to equal latency within a class.
- **Small n.** Three configs have n = $2 \cdot 10^4$. The one with the largest deviation was rerun at 10^5 and matches the analytic mean (ratio 0.9992).
- **Tail extrapolation.** The 2^{-30} tail is extrapolated from the geometric form.
- **Hardware.** One laptop, one run.
- **Shortlist cap.** The cap makes GPU savings a lower bound.
- **Research inputs.** Literature quotes and cited hardware rates were compiled with LLM assistance. Every claim carries its source (`kb/claims/all_claims.yaml`), and the claims behind cited signer rates were checked by hand against their sources.

8. Conclusion and artifacts

Grinding behaves exactly as modelled in tries, costs more per attempt than modelled under FIPS 205, and is a minor part of signing. The signer's hardware budget is a first-order size lever: hundreds of bytes per input. The PQ-signature BIP should state its signer assumption explicitly and consider two named parameter profiles.

Artifacts. `github.com/nanos007/grindstone-pq`: NVRTC kernels, SHA-NI grinder, Python reference, campaign runner, notebooks, `python -m runner.reproduce`. Data under CC-BY-4.0, dataset SHA-256 `01e9fd2ca78f89e5c2525072705e8f32f42863054ffcbc7301341905bfe98d80`.

References

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